SPECIALIZATION OF THE CYCLE OF A FLAT CLOSED SUBSCHEME OVER A DVR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Specialization maps on cycle modules over DVRs (relative-cycles chapter, Lemma lemma-specialization-module, canonical tag 0H4I). cited at proof lines 470-471. Supplies dim(W_kappa) <= r and sp([W_K]_r) = [W_kappa]_r for the module construction with F = O_W.
- Cycle of a closed subscheme equals the cycle of its structure sheaf (Chow Homology, Lemma chow-lemma-cycle-closed-coherent, canonical tag 02QY). cited at proof line 472. Sews the closed subscheme cycle [W] to the module construction with F = O_W.

Source reference. Original lemma 0H4I: sources/stacks-chow/tags.txt, source lines 21353–21353. Cited at proof lines 470-471: supplies the specialization statement for F = O_W.

Source reference. Original lemma 02QY: sources/stacks-chow/tags.txt, source lines 3605–3605. Cited at proof line 472: sews the closed subscheme cycle to the module construction.

Lemma 0.2. *Let R be a discrete valuation ring with fraction field K and residue field κ. Let X be a scheme locally of finite type over R. Let $r \geq 0$. Let $W \subset X$ be a closed subscheme flat over R. Assume* $\dim(W_K) \leq r$. *Then* $\dim(W_\kappa) \leq r$ *and*

$$sp_{X/R}([W_K]_r) = [W_\kappa]_r$$

Proof. Taking $\mathcal{F} = \mathcal{O}_W$ this is a special case of Lemma 0H4I. See Chow Homology, Lemma 02QY. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.